INCH-POUND

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23 February 1989
SUPERSEDING
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27 September 1983

MILITARY SPECIFICATION

CONTAINERS, CARGO

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers nominal 20-foot reusable cargo containers for the transportation, distribution, and storage of military supplies.

1.2 Classification. Containers shall be of the following types, as specified (see 6.2).

Type I - 8'-0" high ISO-1C without mechanical restraint system.
Type II - 8'-0" high ISO-1C with mechanical restraint system.
Type III - 8'-6" high ISO-1CC without mechanical restraint system.
Type IV - 8'6" high ISO-1CC with mechanical restraint system.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-1558 - Paint, Stencil.
NN-P-530 - Plywood, Flat Panel.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8115
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MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-C-46168	- Coating Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-C-62218	- Corrosion Preventive Compounds, Cold-Application (for New Fielded Motor Vehicles and Trailers).

STANDARDS

MILITARY

MIL-STD-889	- Dissimilar Metals.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN; NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

DEPARIMENT OF TRANSPORTATION (DOT)

Bureau of Explosives.
BOE-6000 - Hazardous Materials Regulations of the Department of Transporation (DOT).

Code of Federal Regulations.
CFR 49 - Transportation, Parts 450 and 451.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

INTER-GOVERNMENTAL MARTIME ORGANIZATION

International Convention for Safe Containers (SCS)

(Applicatioon for copies should be addressed to the Commandant, (G-MVI), US Coast Guard, 2100 Second Street, SW Washington, DC 20593.)

DRAWINGS

ME

13219E0037	- Rail, Horizontal.
13219E0038	- Rail, Vertical.
13219E0039	- Beam, Shoring.
13219E0080	- Wood Plank, Floor, Intermediate.

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(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-FMR, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding and Brazing Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 242 - High Strength Low-Alloy Structural Steel.
A 606 - Steel Sheet and Strip, Hot Rolled and Cold-Rolled, High Strength, Low-Alloy, with Improved Corrosion Resistance.
A 847 - Cold-Formed Welded and Seamless High Strength, Low Alloy Structural Tubing with Improved Atmospheric Corrosion Resistance.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

AMERICAN WELDING SOCIETY

D1.1 - Structural Welding Code - Steel.
D1.3 - Structural Welding Code - Sheet Metal.

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJune Road, P. O. Box 351040, Miami, FL 33126.)

ASSOCIATION OF AMERICAN RAILROADS (AAR)

Rules Governing the Loading of Commodities on Open-Top Cars.

(Application for copies should be addressed to the Association of American Railroads, American Railroads Building, 50 F Street, NW, Washington, DC 20001.)

INTERNATIONAL STANDARDS ORGANIZATION (ISO)

ISO 668 - Series 1 Freight Containers - Classification External Dimensions and Ratings.
ISO 830 - Terminology Relating to Freight Containers.
ISO 1161 - Specification of Corner Fittings for Series 1 Freight Containers.

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ISO 1496/1 - Series 1 Freight Containers - Specification and Testing - Part 1: General Cargo Containers for General Purposes.
ISO 6346 - Freight Containers - Coding, Identification and Marking.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The containers shall be noncollapsible of a permanent character and suitable for repeated use.

3.1.1 Terms and definitions. The terms used in this specification shall be as defined in ISO 830.

3.1.2 Weight, ratings, and dimensions. The tare weight of the container shall be of the minimum practical, but shall not exceed 5,000 pounds for type I, 5,940 pounds for type II, 5,200 pounds for type III, and 6,140 pounds for type IV. The gross weight ratings for each container shall be 20 long tons or 44,800 pounds. Dimensions, tolerances, and diagonal differences of the container shall meet the requirements for 20-foot nominal length containers as specified in ISO 668. All inside dimensions shall be the maximum attainable, consistent with design and material, but not less than those shown in table I. Door opening dimensions shall be in accordance with ISO 1496/1. When forklift pockets are furnished, the maximum permitted tare weight of the container shall be increased by 150 pounds.

TABLE I. Dimensions.

Item	Dimension
Inside width, minimum.	91-3/4 inches (type I & III); 91-1/4 inches (type II & IV).
Inside length, minimum.	231 inches.
Inside height, minimum (top of floor to underside of roof bows).	86-1/2 inches (type I & II); 92-1/2 inches (type III & IV).

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3.1.2.1 Tare weight. The tare weight to be marked on the container (see 3.6) shall be the average weight of the first 10 production units. The tare weight of the remaining production containers shall be within plus or minus 35 pounds of the stamped weight.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. Materials shall be as specified herein. Materials not definitely specified shall be in accordance with a Federal, Military or National Technical Society, Association or Institute specification or standard.

3.3.1 Steel requirement. The container shall be constructed of high-strength low alloy (HSLA) structural and steel conforming to ASTM A 242, type I or II, HSLA structural tubing conforming to ASTM A 847, and sheet steels conforming to ASTM A 606, type 4.

3.3.2 Material deterioration prevention and control. The container shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the container may be exposed.

3.3.3 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.4 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.5 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the container may be newly fabricated from recovered materials to the maximum extent practicable, provided the container produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the container.

3.4 Construction. Containers shall be constructed so as to be free of any recesses and voids in which contraband can be concealed or where moisture can accumulate. No part of the container (when empty) shall protrude beyond the outside surfaces of the corner fittings. The yield point of materials shall not be exceeded when the container is tested as specified in 4.5.2. Material for sides, ends and top shall be steel.

3.4.1 Side walls and front wall. The steel side walls and front wall may be of the interior or exterior post type, corrugated, or of smooth skin construction. On all types of construction, lap splices are permitted on metal skin. The interiors of the metal sides and front walls of type I and III containers shall be lined, full height, with 1/4-inch thick plywood conforming to

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NN-P-530, group B, grade B-C or C-C (plugged), exterior type. The interior of the front walls of type II and IV containers shall be lined, full height, with 1/4-inch thick plywood conforming to NN-P-530, group B, grade B-C or C-C (plugged), exterior type. The better surface shall face inside the container. All plywood joints and ends shall be fastened or bonded to the container. A kick plate 12 inches high and made of 18-gage metal shall be attached to the bottom of the interior sides and front end wall over the plywood liner of type I and III containers. Type II and IV containers shall have the kick plate attached to the bottom of the front end wall of the plywood liner. The kick plate shall be fully bonded to the plywood.

3.4.2 Floor construction. The container floor shall be fabricated of laminated or solid hardwood boards conforming to drawing 13219E0080. The boards shall run longitudinally and be of the same width and thickness. The board edges shall be jointed by ship-lap construction. The boards shall be attached to the cross members by means of at least three fasteners per board, per crossmember, for boards over 7 inches wide; and two fasteners per board, per crossmember, for boards less than 7 inches wide. Countersunk head fasteners, 1/4 inch diameter minimum, either of the self-tapping screw type, or machine screws with self-locking nuts, shall be installed so that each head is 0 to 1/16 inch below the board surface and at least one inch from the board edge. The floor shall be watertight. The floor shall be installed to permit lateral variations in floorboard width due to swelling.

3.4.3 Understructure. The container bottom side rails and bottom end frame members shall not deflect more than 1/4 inch below the bottom plane of the bottom corner fittings when the container is tested as specified in 4.5.2. Crossmembers shall not deflect below the bottom plane of the bottom corner fitting when the container is tested as specified in 4.5.2; except that when tested as specified in 4.5.2.2.8, they may deflect to not more than 1/4 inch below the bottom plane of the bottom corner fittings. All cross-members shall be of the same configuration and strength and shall have a maximum center-to-center distance of 12 inches, except crossmember supports within 24 inches of the threshold shall have a minimum center-to-center distance of 8 inches.

3.4.3.1 Forklift pockets. When specified (see 6.2), forklift pockets for handling unloaded containers shall be furnished. Forklift pockets shall conform to ISO 1496/1.

3.4.4 Roof construction. The roof shall be self draining. Lap splices are permitted. Corrugated roofing construction is permitted. If the roof bows are fastened to the roof sheet, they shall be bonded by a suitable bonding agent. In the case of roof bows that are not fastened to the roof sheet, anti-chafing material shall be affixed to the roof bows on the surfaces facing the roof sheets. The roof sheet, roof rails, and upper end frame members shall be at least 1/4 inch below the top plane of the top corner fittings.

3.4.5 Doors. Two doors shall be hung within the rear end frame and shall provide a clear opening conforming to the dimensions specified in 3.1.2. Heavy duty pin hinges, recessed within the corner structure, shall be provided on each door allowing the door to fold back against the sides of the body. Steel hinges shall have corrosion-resistant steel pins. Each door shall be provided with one or more heavy-duty, handle-operated cam locking devices with anti-rack

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provisions which through lever type action aid in releasing the door seal from the door frame. Handles shall be 12 to 18 inches above the bottom plane of the bottom corner fittings. Each locking device handle must be capable to accept a padlock or security seal. Means shall be provided to hold the doors in the full open position. The means provided to hold the doors in the open position shall be of a material which will not scrape or chafe the container when the doors are closed. The doors, when closed, shall be sealed in such a manner as to prevent water entry into the container. All moving parts of the door locking mechanism and door hinges, if not permanently lubricated, shall be lubricated with grease conforming to MIL-G-10924.

3.4.6 Corner fittings. The corner fittings shall conform to the requirements of ISO 1161.

3.4.7 Anti-pilferage provisions. Hinge-pins and screws, bolts, and other fasteners used for securing the hinges and closing devices to the container and for holding the essential parts of the sides, ends, and roof, shall be tack welded or otherwise secured in such a manner as to prevent access to the interior of the container without leaving visible signs of tampering. Where such welding destroys protective coating on the items being welded or on other container parts, the weld and surrounding area shall be thoroughly cleaned, treated, and painted in accordance with 3.5. All locking device handles shall be furnished with provisions for padlocking and customs sealing.

3.4.8 Restraint system. Type II and IV containers shall be furnished with ammunition restraint systems. The internal cargo restraint system shall be capable of accepting the shoring beams conforming to drawing 13219E0039. The shoring beams shall be approved for use in the shipment of ammunition in containers by the Bureau of Explosives of the Association of American Railroads (AAR), the US Coast Guard, and the US Army Defense Ammunition Center and School. The restraint system shall consist of eight horizontal rails conforming to Drawing 13219E0037 attached to the walls of the container, a vertical rail conforming to Drawing 13219E0038 attached on each side of the door frame, and 25 shoring beams. The horizontal rails on the side walls shall be located so that the top surfaces of shoring beams, when attached to the side rails, shall be 5, 16, 28, 38, 48, 60, 72, and 83 inches, respectively, above the floor. The vertical rails in the door frame shall provide for location of the shoring beams at approximately the same heights as provided by the side rails.

3.5 Cleaning, treatment and painting. Metal surfaces of the container shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. After cleaning, treatment and primer, all exterior metal surfaces shall be painted in accordance with MIL-C-46168, color camouflage green 383, unless otherwise specified (see 6.2). When specified (see 6.2), the container shall be overcoated in accordance with the Government furnished camouflage patterns with MIL-C-46168 top coat of the colors specified in the camouflage patterns except the first article may be camouflage green 383. The shoring beams of the mechanical restraint system shall not be painted.

3.5.1 Wood surfaces. Prior to undercoating, all wood surfaces shall be cleaned in accordance with MIL-T-704.

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3.5.1.1 Plywood. Treatment and painting of plywood members is not required.

3.5.2 Undercoating. The entire underside of the container floor, including crossmembers, corner fittings, side rails, and end frame members, shall be coated after painting of the metal surfaces, with a type I coating conforming to MIL-C-62218.

3.6 Identification marking. The container shall be identified and marked in accordance with ISO 6346 with the following additional requirements:

a. The following data shall be marked on the right-hand door in characters not less than 2 inches high:

(1) Maximum gross weight in pounds.
(2) Tare weight in pounds.
(3) Net cargo weight in kilograms, tons, and pounds.
(4) Shipping cube in cubic meters and cubic feet.
(5) Nominal outside dimensions in meters and feet.

b. The owner code and the serial number shall be as specified (see 6.2). All markings shall be white stencil paint conforming to A-A-1558. Spacing between letters shall be even to present a neat, readable appearance.

c. When the containers are furnished without forklift pockets, the following warning shall be marked on both sides at the base line, centrally located, "Caution do not handle container with forks", letter size to be a minimum of 3 inches. When forklift pockets are furnished, the following warning shall be marked on both sides, centrally located, "FOR EMPTY LIFT ONLY", letter size to be a minimum of 3 inches.

3.6.1 Interior marking. The container control number shall be stamped or bead welded in characters not less than 1/2 inch high on an interior surface of the rear upper end frame member. The number shall be located within an area 18 inches from the left vertical end frame member where it will not be obscured.

3.6.1.1 Marking for stowage of shoring beams, type II and type IV. Each type II and type IV container, equipped with a belt rail cargo restraint system, shall be stenciled on the inside surface of the right hand door with white letters 1-1/4 inches high as follows: "25 shoring beam assemblies shall remain with the container."

3.6.2 Data plates. A nonferrous metal data plate conforming to MIL-P-514, type I, style 3, composition C shall be affixed to the center of the external surface of the right hand-door. The method of attachment shall be by either riveting or bolting. The data plate shall contain the following information:

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(NAME OF OWNER, see 6.2)
CONTAINER, CARGO
SPECIFICATION: MIL-C-52661
NSN:_____
CONTROL NUMBER:_____
TARE WEIGHT: LBS_____KG_____
CONTRACT NUMBER_____
MFD BY:_____
DATE (MONTH & YEAR):_____
TECHNICAL MANUAL:_____

3.6.3 CSC plate. A Convention for Safe Containers (CSC) plate shall be applied for and obtained from a designated approval authority, attached and displayed as required by the convention in accordance with CFR 49, parts 450 and 451.

3.7 Placard holder. Four stainless steel placard holders conforming to DOT, BOE-6000, part 172, appendix C, shall be provided. One shall be located on each end and each side of the container and be positioned with the top of the holder approximately 2 feet above the bottom plane of the bottom corner fittings. Holders shall be at least 3 inches from the other container markings. If the container design permits, the holders shall be positioned within a recessed area of the structure to preclude the holders from being damaged during container handling and transport. Placard holders shall be permanently attached, welded when applicable, in the designated locations.

3.8 Performance. The container shall conform to the requirements specified herein under all conditions specified without damage or permanent deformation. There shall be no evidence of accelerated wear or failures when tested under all conditions specified herein.

3.8.1 Stacking. Stacking capability shall be in accordance with ISO 1496/1.

3.8.2 Lifting from the top. Lifting from the top capability shall be in accordance with ISO 1496/1.

3.8.3 Lifting from the bottom. Lifting from the bottom capability shall be in accordance with ISO 1496/1.

3.8.4 Longitudinal restraint. Restraint capability shall be in accordance with ISO 1496/1.

3.8.5 Floor. The floor of the container shall withstand loading in accordance with ISO 1496/1.

3.8.6 Roof. The container roof shall withstand loading in accordance with ISO 1496/I.

3.8.7 Walls. Each of the end and side walls shall withstand loading in accordance with ISO 1496/1.

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3.8.8 Transverse and longitudinal rigidity. The container shall withstand racking loads in accordance with ISO 1496/1 with the additional requirement that the door end of the container shall show no signs of water leakage during the test specified herein.

3.8.9 Weatherproofing. The container shall show no signs of leakage when tested in accordance with 4.5.2.2.10.

3.8.10 Rail impact test. The container shall exhibit none of the failure criteria listed in 4.5.2.3 after completion of the rail impact test as specified in 4.5.2.2.12.

3.9 Workmanship. All parts, components, and assemblies of the container including castings, forgings, molded parts, stampings, seals and sealing agents, machined surfaces, and welded parts shall be clean and free from any defects that will reduce the capability of the container to meet the requirements specified herein. Any components and assemblies which have been repaired or modified to overcome deficiencies shall not be used without prior specific approval of the contracting officer. External surfaces shall be free from burrs, slag, sharp edges, and corners except where sharp edges and corners are required. The internal cargo space shall be free from sharp protrusions that could damage cargo or personnel.

3.9.1 Metal fabrication. Metal used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and clipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting, using a tip suitable for the thickness of the metal, may be employed instead of shearing or sawing. Burned surfaces of flame-cut material shall be free of slag. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly.

3.9.2 Bolted and riveted connections. Bolted and rivet holes shall be accurately punched or drilled and shall have the burrs removed. Washers, lockwashers, or lock nuts shall be provided where necessary and all bolts, nuts, and screws shall be tight. Rivet heads, when not countersunk or flattened, shall be of uniform size and shape for the same diameter of rivet, concentric with the rivet holes, and in full contact with the surface of the member.

3.9.3 Welders and welding.

3.9.3.1 Welders or welding operator. Before assigning any welder or welding operator to work covered by this specification, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed by one of the following listed codes for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

AWS D1.1 Structural Welding Code - Steel.
AWS D1.3 Structural Welding Code - Sheet Steel.
ASME Boiler and Pressure Vessel Code, Section IX, Welding and Brazing.

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Contractors who only make horizontal welds need not qualify welders or welding operators for "all position welding". In the event of evidence of poor welds, the Government reserves the right to require retesting of any welder or welding operator. All certification test results shall be made available for review by the contracting officer or the contracting officer's representative.

3.9.3.2 Welding. All welding and welding procedures shall be in accordance with the applicable procedures as defined by the following codes:

AWS D1.1 Structural Welding Code - Steel.
AWS D1.3 Structural Welding Code - Sheet Steel.
ASME Boiler and Pressure Vessel Code, Section IX, Welding and Brazing.

The surface of parts to be welded shall be free from rust, scale, oxides, paint, grease, or other foreign matter. Weldments shall be of homogeneous appearance and form without spattering or other irregularities, complying with the size, configuration, any other dimensional requirements to develop the full strength of the parts joined by the welds. No cracks of any type are permitted. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings. All welds shall be inspected and tested in accordance wtih the criteria for workmanship and welding quality as stated in the applicable codes.

3.9.4 Machined work. All parts shall be manufactured to gage through the use of jigs and fixtures and shall provide interchangeability of parts as manufactured.

3.9.5 Castings. Castings shall be sound and free from patching, misplaced coring, warping, or defects which might render the castings unsound for use.

3.10 Government furnished property. When specified (see 6.2), the following property in the quantity indicated will be furnished by the Government (see 6.4).

Item No.	Description	Quantity
1	Camouflage paint pattern	As required

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the

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contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspection. The inspection requirements are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.6).
d. Inspection of packaging (see 4.7).

4.3 First article inspection.

4.3.1 In-process inspection. The two first-produced containers shall receive an in-process inspection by the contractor under the direction and surveillance of the Government. This inspection shall be conducted to evaluate materials and workmanship as specified in 3.3 and 3.9. Nonconformance to 3.3 or 3.9 shall be cause for rejection.

4.3.2 Examination. The two first-produced containers shall be examined as specified in 4.4.1 and 4.5.1. Presence of one or more defects in either container shall be cause for rejection of both containers.

4.3.3 Tests. The two first-produced containers shall be tested as specified in 4.5.2 through 4.5.2.2.12. Tests specified for single containers shall be performed on both containers. Failure of any test shall be cause for rejection of both containers and shall be cause for the Government to defer future acceptance until objective evidence furnished by the contractor indicates that deficiencies revealed by the tests have been corrected.

4.4 Quality conformance inspection.

4.4.1 Examination. Each container shall be examined for defects listed in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each container shall be tested for weatherproofness in accordance with 4.5.2.2.10 before application of finish painting or undercoating, or installation of plywood liner. Penetration of moisture shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The containers shall be examined for the following defects:

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Major

101. Dimensions not as specified. Each incorrect dimension shall be considered one defect.
102. Components missing or not as specified. Each missing component shall be considered one defect.
103. Assembly incomplete or not as specified.
104. Doors do not open, close, or latch when operated by one man standing on the ground and without the use of tools.
105. Door seals not as specified.
106. Tare weight in excess of that specified (see 3.1.2).
107. Tare weight not within specified limits (see 3.1.2.1).
108. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration (see 3.3.2).
109. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other(see 3.3.3).
110. Contractor does not have documentation available for identification of materials, material finishes, or treatment (see 3.3.4).
111. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the container (see 3.3.5).
112. Construction not as specified (see 3.4).
113. Side and front walls not as specified (see 3.4.1).
114. Floor construction not as specified (see 3.4.2).
115. Understructure not as specified (see 3.4.3).
116. Forklift pockets not as specified, when furnished (see 3.4.3.1).
117. Roof construction not as specified (see 3.4.4).
118. Doors not as specified (see 3.4.5).
119. Corner fittings not as specified (see 3.4.6).
120. Restraint system not as specified (see 3.4.9).
121. Cleaning, treatment and painting not as specified (see 3.5).
122. Color not as specified (see 3.5).
123. Wood surfaces not cleaned as specified (see 3.5.1).
124. Underside of container floor not coated as specified (see 3.5.2).
125. Data plates not as specified (see 3.6.2).
126. CSC plates not as specified (see 3.6.3).
127. Placard holder not as specified (see 3.7).
128. Workmanship not as specified (see 3.9).
129. Metal fabrication not as specified (see 3.9.1).
130. Fasteners missing (see 3.9.2).
131. Fasteners not countersunk as specified (see 3.9.2).
132. Welding and welding specifications not as specified (see 3.9.3).
133. Welder's certification not available (see 3.9.3.1).
134. Machine work not as specified (see 3.9.4).
135. Castings not as specified (see 3.9.5).

Minor

201. Door hinges and latches not lubricated as specified (see 3.4.5).
202. Anti-pilferage provisions not as specified (see 3.4.7).

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4.5.2 Tests.

4.5.2.1 Test conditions. The container shall first be tested for weatherproofness before application of finish paint or undercoating or installation of plywood liner. Other tests shall be made after application of finish painting and undercoating and installation of plywood liner (if required). A second test for weatherproofness shall be made on the doors during racking tests of the door end. A final test for weatherproofness shall be made after completion of all other tests. The test load within the container shall be uniformly distributed, except as otherwise specified herein. On completion of each test, the container shall be examined to determine compliance with the performance requirements specified in 3.8.

4.5.2.2 Test procedures.

4.5.2.2.1 Stacking. A stacking test shall be conducted in accordance with ISO 1496/1.

4.5.2.2.2 Lifting from the top. Top lifting capability shall be tested in accordance with ISO 1496/1.

4.5.2.2.3 Lifting from the bottom. Bottom lifting capability shall be tested in accordance with ISO 1496/1, with the additional requirement that the line of action of the lifting cables and the outer face of the corner fittings shall be parallel and not more than 1-1/2 inches apart.

4.5.2.2.4 Longitudinal restraint. Restraint capability shall be tested in accordance with ISO 1496/1, with the additional requirements that each side may be tested individually using a force of 56,000 pounds in tension and compression.

4.5.2.2.5 End wall strength. The end wall strength shall be tested in accordance with ISO 1496/1. The container shall be held in the test load position for a period of not less than 5 minutes.

4.5.2.2.6 Side wall strength. The side wall strength shall be tested in accordance with ISO 1496/1. The container shall be held in the load position for a period of not less than 5 minutes.

4.5.2.2.7 Roof strength. Roof strength shall be tested in accordance with ISO 1496/1 with the additional requirement that test loading shall be applied for not less than 5 minutes.

4.5.2.2.8 Floor strength. Floor strength shall be tested in accordance with ISO 1496/1. Tests shall be made with the container resting on its four bottom corner fittings, with the intermediate structure not less than 1 inch above the supporting surface and free to deflect.

4.5.2.2.9 Forklift pockets. When furnished (see 6.2), the forklift pockets shall be tested in accordance with ISO 1496/1.

4.5.2.2.10 Weatherproofness. Each container shall be tested for weatherproofness before application of finish painting or undercoating or installation of the plywood liner. A stream of water shall be applied to all joints, seams and

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gasketed closures (except during racking tests) of the container from an 0.5 inch inside diameter nozzle at a nozzle pressure of not less than 15 psig. Rate of movement over the exterior of the container shall be approximately 4 feet per second. The nozzle shall be held 5 feet away from the container under test. Tests shall be continued for not less than 15 minutes (except during rigidity tests). On completion of the test the container shall be examined for penetration of water into the interior of the container. Penetration of moisture shall constitute failure of this test.

4.5.2.2.11 Transverse and longitudinal rigidity. Test shall be conducted in accordance with ISO 1496/1 with additional requirements that, (1) both doors shall be closed and latched during test, (2) racking forces shall be applied first laterally and then longitudinally to each top corner fitting in turn, (3) each force application shall be for not less than 5 minutes, and (4) the container shall be tested for weatherproofness during each force application to a top rear corner fitting. The weatherproofness test shall conform to 4.5.2.2.10, except that the water shall be applied to the door end of the container for not less than 5 minutes.

4.5.2.2.12 Rail impact test.

Step 1. From one to five railcars shall be positioned on a level section of track. The following conditions shall be satisfied:

(1) The total weight of the car(s) shall be at least 114,000 kilograms (250,000 pounds).
(2) The coupler between any cars shall be compressed to take up any slack.
(3) All of the air and hand brakes of the car(s) used shall be set.
(4) Any load in or on the car(s) shall be secured to prevent sliding or shifting; any movement greater than 5 centimeters (1.97 inches) shall be justification for retest.
(5) The end of the buffer car to be struck must have a standard draft gear (non cushioned coupler).

Step 2. Mount the test item on the test car. The test car shall be equipped with standard draft gears and conventional under frames. Mounting of the test item shall incorporate the standard loading and bracing method as shown in section 6 of the Association of American Railroads (AAR) "Rules Governing the Loading of Department of Defense Material on Open Top Cars." No exotic or unusual tiedown methods shall be used; any nonstandard loading and bracing must be approved by the Military Traffic and Management Command Transportation Engineering Agency (MTMCTEA), ATTN: MTT-TR, P. O. Box 6276, Newport News, VA 23606-0276 prior to testing. Container types II and IV shall be fully loaded using test loads of equal weight, weight distribution and similar in shape to class "A" explosives. The load shall be secured using the mechanical restraint devices. The arrangement of the test item and its blocking and tiedown to be tested shall be identical to that proposed and approved by MTMCTEA (if nonstandard).

This requirement is mandatory for all equipment developed for use in the U.S. Equivalent European standards may apply for non-U.S. commodities.

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Step 3. Situate the test car between the buffer cars and the locomotive, and pull the test car at least 65 meters (200 feet) from the buffer car(s) along a level section of track (a minimum distance to achieve the required locomotive speeds).

Step 4. Position the knuckles of the buffer and test cars for coupling.

Step 5. Install a timing device to measure the test car speed (±0.1 km/hr) just prior to impact with the buffer car(s). Suggested methods include electronic timing (microswitches) and radar. The use of torpedoes and a stopwatch is permissible but not recommended because of the inaccuracies involved.

Step 6. Push the test car toward the buffer car(s) and, by using the locomotive's speedometer or other means, release the test car to freely impact the buffer car(s). The door end of the container shall face in the direction of the buffer car.

Step 7. Repeat Step 6 until the test car has impacted the buffer car(s) three times at the same end, once each at speeds of 6.4, 9.7, and 13.0 km/hr (4, 6, and 8 mph) +0.8, -0.0 km/hr. Reverse the test car and repeat the 13.0 km/hr impact, for a total of 4 impacts.

NOTE: Adjustment of the loading or securing mechanisms, or reconditioning of the bracing or items of securement is not allowed without a complete retest.

Step 8. Record the pertinent information for each impact, to include the following:

a. Complete test item identification.
b. The number and speed of impact.
c. Observations of tiedowns, blockings, fittings.
d. Observations of the test item's physical condition.
e. Results of any operational checks.

4.5.2.2.13 CSC approval testing. Convention of SAFE Containers (CSC) approval is required. Additional testing requested by the certifying agency shall be conducted prior to CSC approval.

4.5.2.3 Failure criteria. Any of the following shall constitute failure of any of the tests specified in 4.5.2. After removal of the load, and before checking criteria of "d", "h", and "j", the container under test shall be subjected to shock and vibration to eliminate temporary residual bulge.

a. Deflection of the container bottom side rails or bottom end frame members of more than 1/4 inch below the bottom plane of the bottom corner fittings.
b. Deflection of the container crossmember below the limits specified in 3.4.3.
c. Evidence of weld failure.
d. Permanent deformation.
e. Any broken or torn component.
f. Any cracked or broken floorboard.

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g. Any detachment of a gasket.
h. Inability of one man standing on the ground to open or close a door without the use of tools.
i. Inability of one man standing on the ground to operate a latching mechanism without the use of tools.
j. Any dimension after a test not within the specified tolerances (dimension changes which persist for 2 hours after removal of the load and subjecting the container to shock and vibration.
k. Penetration of water into the interior of the container.
l. Inability to obtain CSC approval.

4.6 Inspection comparison. The Government may select containers at any time during the contract production period and subject these containers to the examinations specified in 4.5.1 and the tests specified in 4.5.2 (except for the first weatherproofness test specified in 4.5.2.1) to determine conformance to the requirements of this specification. This inspection will be performed by the Government at a site selected by the Government on containers selected at random from those which have been accepted by the Government and will not include the previously inspected first-produced containers. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as a part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison container to meet any requirement specified herein during and as a result of the examinations and tests specified in 4.6 shall be cause for rejection of the inspection comparison container and shall be cause for refusal by the Government to continue acceptance of production containers until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on containers previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all containers accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging.

4.7.1 Quality conformance inspection of pack.

4.7.1.1 Unit of product. For the purpose of inspection, a container prepared for shipment shall constitute a unit of product.

4.7.1.2 Examination. Each container shall be examined for the following defects. Presence of one or more defect shall be cause for rejection.

137. Markings missing, illegible, or incorrect (see 3.6 thru 3.6.1.1).

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5. PACKAGING

5.1 Preservation and packing. Preservation and packing is not applicable for ISO cargo containers.

5.2 Marking. Destination markings for shipment from the contractor's plant to the initial receiving activity shall be marked on tags with waterproof ink and the tags securely fastened to the container door handle hardware. Other required contract markings can be found on the container data plate.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The reusable cargo container is intended for transporting military and quasi-military cargo. The container can be lifted from the top or bottom corner fittings, and can be carried by highway, railroad, water, and air modes of transportation.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number, and date of the specification.
b. Type of container required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2..1.1 and 2.2).
d. When the Government will conduct any or all of the first article container examinations and tests. When the Government will conduct a portion of the first article examinations and tests, the contracting officer should specify which examinations and tests will be conducted by the Government and which examinations and tests will be conducted by the contractor (see 3.2).
e. Time frame required for submission of first article containers (see 3.2).
f. When forklift pockets are required (see 3.4.3.1).
g. When camouflage pattern overcoating will be provided (see 3.5 and 6.4).
h. Color when other than as specified (see 3.5).
i. Owners code and serial number for each container (see 3.6).
j. Name of owner to be inscribed on data plate (see 3.6.2).

6.3 First article. When a first article inspection is required, the item(s) should be a first-produced model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.4 Government furnished property. The contracting officer shall arrange to furnish the property specified in 3.10.

6.5 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.7 Subject term (key word) listing.

Cargo container
Container, 20 foot
Container, cargo
Container, reusable

Custodian:
Army - ME

Preparing activity:
Army - ME

Review activities:
Army - GL, MT

Project 8115-A506

User activity:
Army - SM

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-52661C(ME)	Containers, Cargo

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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